EDA MINI PROJECT

EnTC A-3 | Group 1 | Semester 2 – B.Tech

Analysis on Student Daily Commute

EDA Questions & Analysis Index

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Dataset: Google Form Survey (724 responses)

Subject: Exploratory Data Analysis

> **About this document:** This document lists every analytical question investigated in our EDA notebook — organised by technique. For each question you will find the exact Python method used, so this sheet can serve as a quick reference alongside the notebook.

SECTION A – DATASET LOADING & BASIC INSPECTION

The first step of any EDA is to load the dataset and understand its basic structure — how many rows and columns there are, what data types exist, and what the first and last few records look like.

#	Question / Analysis Performed	Python Technique
Q1	How do we load the student commute dataset into a pandas DataFrame?	pd.read_csv()
Q2	What is the total number of records (rows) and features (columns) in the dataset?	df.shape
Q3	What are the first 5 entries in the dataset?	df.head()
Q4	What are the last 5 entries in the dataset?	df.tail()
Q5	What are the data types of each column in the dataset?	df.dtypes
Q6	What is the general structure and memory usage of the dataset?	df.info()
Q7	What are the names of all columns in the dataset?	df.columns

SECTION B – DESCRIPTIVE STATISTICS

Descriptive statistics summarise the central tendency, spread, and shape of numerical data. We applied these to the two key numeric columns: Travel_Time_min and One_Way_Cost_INR.

#	Question / Analysis Performed	Python Technique
Q8	What are the basic statistical measures (count, mean, std, min, max, quartiles) for all numerical columns?	df.describe()
Q9	What is the average one-way travel time (in minutes) of students?	df['Travel_Time_min'].mean()
Q10	What is the median travel time, and how does it compare to the mean?	df['Travel_Time_min'].median()
Q11	How much variation is there in student travel times? (Standard Deviation)	round(df['Travel_Time_min'].std(), 2)
Q12	What is the shortest and longest recorded one-way travel time?	df['Travel_Time_min'].min() / .max()
Q13	What is the average one-way commute cost (in ₹) across all students?	df['One_Way_Cost_INR'].mean()
Q14	What is the median commute cost, and is the cost distribution skewed?	df['One_Way_Cost_INR'].median()
Q15	What is the cheapest and most expensive one-way commute recorded?	df['One_Way_Cost_INR'].min() / .max()

SECTION C – MISSING VALUE ANALYSIS

Before performing analysis it is critical to understand where data is missing and decide how to handle it. 132 values were found missing in the Delay_Points column and were filled with 'None'.

#	Question / Analysis Performed	Python Technique
Q16	Are there any missing (null) values in the dataset? If yes, in which columns?	df.isnull().sum()
Q17	What percentage of data is missing in each column?	df.isnull().sum() / len(df) * 100
Q18	How should we handle the 132 missing values in the Delay_Points column?	df['Delay_Points'].fillna('None')
Q19	After filling missing values, is the dataset completely clean (zero nulls)?	df.isnull().sum() (post-fill)

SECTION D – FREQUENCY COUNT

Frequency counts tell us how often each category appears in the data. We applied this across all categorical columns to understand distribution at a glance.

#	Question / Analysis Performed	Python Technique
Q20	Which primary mode of transport is used the most by students?	df['Primary_Mode'].value_counts()
Q21	During which time slot do most students commute?	df['Time_of_Day'].value_counts()
Q22	Which area contributes the highest number of commuting students (starting point)?	df['Starting_Point'].value_counts()
Q23	Which destination is most frequently travelled to?	df['Destination'].value_counts()
Q24	What is the most common delay/bottleneck faced by students on their commute?	df['Delay_Points'].value_counts()
Q25	How are satisfaction ratings distributed across all responses?	df['Satisfaction'].value_counts().sort_index()

SECTION E – UNIQUE VALUES

Identifying unique values helps us understand the range of categories in each column and detect any unexpected entries.

#	Question / Analysis Performed	Python Technique
Q26	What are all the distinct starting points (areas) present in the dataset?	df['Starting_Point'].unique()
Q27	What are all the distinct destinations students commute to?	df['Destination'].unique()
Q28	How many different primary modes of transport are used by students?	df['Primary_Mode'].unique()
Q29	What are the unique time slots in which students commute?	df['Time_of_Day'].unique()
Q30	What are the unique delay points/bottlenecks reported across all commutes?	df['Delay_Points'].unique()

#	Question / Analysis Performed	Python Technique
Q31	How many unique values exist in each column of the dataset?	df.nunique()

SECTION F – PERCENTAGE DISTRIBUTION

Percentage distributions allow us to compare categories on a common scale, regardless of total count. This is especially useful when comparing across groups.

#	Question / Analysis Performed	Python Technique
Q32	What percentage of students use each primary mode of transport?	value_counts(normalize=True) * 100
Q33	What percentage of commutes happen in each time slot of the day?	value_counts(normalize=True) * 100
Q34	What is the percentage share of each satisfaction rating (1 to 5)?	value_counts(normalize=True) * 100
Q35	What percentage of students face each type of delay/bottleneck?	value_counts(normalize=True) * 100

SECTION G – FILTERING DATA

Filtering allows us to isolate subsets of data to answer specific targeted questions about a particular group, route, or condition.

#	Question / Analysis Performed	Python Technique
Q36	Which students use Metro as their primary mode of transport? Show their full commute details.	df[df['Primary_Mode'] == 'Metro']
Q37	Which commutes take place specifically during the Morning Peak (8 AM – 11 AM)?	df[df['Time_of_Day'] == 'Morning Peak...']
Q38	Which commutes cost more than ₹300 one-way? Which modes are responsible?	df[df['One_Way_Cost_INR'] > 300]
Q39	Which commutes take more than 90 minutes? What routes and modes are involved?	df[df['Travel_Time_min'] > 90]
Q40	What are the commute details specifically for students starting from Kothrud?	df[df['Starting_Point'] == 'Kothrud']

SECTION H – CROSS TABULATION (PD.CROSSTAB)

Cross tabulation produces frequency tables that show the relationship between two categorical variables. It reveals patterns and combinations that individual frequency counts miss.

#	Question / Analysis Performed	Python Technique
Q41	What is the distribution of primary modes of transport across different time slots of the day?	pd.crosstab(Primary_Mode, Time_of_Day)
Q42	Which mode of transport is predominantly used from each starting point/area?	pd.crosstab(Starting_Point, Primary_Mode)
Q43	What modes of transport are used to reach each destination?	pd.crosstab(Destination, Primary_Mode)
Q44	How does satisfaction rating vary across different primary modes of transport?	pd.crosstab(Primary_Mode, Satisfaction)
Q45	During which time slots does each type of delay/bottleneck occur most frequently?	pd.crosstab(Time_of_Day, Delay_Points)

SECTION I – GROUPBY ANALYSIS

Groupby allows us to compute aggregate statistics (mean, count, max, etc.) across each category. This is the core technique for comparing groups within the dataset.

#	Question / Analysis Performed	Python Technique
Q46	What is the average travel time and average cost for each primary mode of transport?	groupby('Primary_Mode')[...].mean()
Q47	How does average travel time and cost differ across different times of the day?	groupby('Time_of_Day')[...].mean()
Q48	Which mode of transport has the highest average satisfaction rating from students?	groupby('Primary_Mode')['Satisfaction'].mean()
Q49	How many students commute from each starting point/area?	groupby('Starting_Point')['Satisfaction'].count()

#	Question / Analysis Performed	Python Technique
Q50	What is the maximum one-way cost recorded for each primary mode of transport?	groupby('Primary_Mode')['One_Way_Cost_INR'].max()

SECTION J – SORTING

Sorting helps identify extreme values and rank records by a particular metric — useful for spotting the fastest, slowest, cheapest, or most expensive commutes.

#	Question / Analysis Performed	Python Technique
Q51	Which 10 commutes have the shortest travel time? What modes are involved?	sort_values('Travel_Time_min').head(10)
Q52	Which 10 commutes are the most expensive (one-way cost)? What modes are used?	sort_values('One_Way_Cost_INR', ascending=False).head(10)
Q53	Which 10 commutes received the highest satisfaction ratings?	sort_values('Satisfaction', ascending=False).head(10)

SECTION K – DERIVED COLUMN (FEATURE ENGINEERING)

A derived column is a new column created from existing data. Here we categorised each commute into Short, Medium, or Long based on travel time — making pattern analysis easier.

#	Question / Analysis Performed	Python Technique
Q54	Can we classify each commute as Short (≤30 min), Medium (31–60 min), or Long (>60 min)?	df.apply() with custom function
Q55	How many commutes fall into each category (Short / Medium / Long)?	value_counts() on Commute_Category
Q56	What percentage of commutes are Short, Medium, and Long?	value_counts(normalize=True) * 100
Q57	For each Commute Category, which primary modes of transport are most commonly used?	pd.crosstab(Commute_Category, Primary_Mode)

SECTION L – VISUALIZATIONS

Visual representations make patterns in the data immediately clear. The following charts were produced as part of the EDA notebook.

#	Question / Analysis Performed	Python Technique
V1	Which primary mode of transport is used most? (Bar Chart)	df['Primary_Mode'].value_counts().plot(kind='bar')
V2	What proportion of commutes happen in each time slot? (Pie Chart)	df['Time_of_Day'].value_counts().plot(kind='pie')
V3	How are travel times spread across all students? (Histogram)	plt.hist(df['Travel_Time_min'], bins=20)
V4	How is commute cost distributed across all students? (Histogram)	plt.hist(df['One_Way_Cost_INR'], bins=20)
V5	Which bottleneck/delay point affects students the most? (Bar Chart)	df['Delay_Points'].value_counts().plot(kind='bar')
V6	Which mode of transport has the fastest average travel time? (Horizontal Bar Chart)	groupby('Primary_Mode')['Travel_Time_min'].mean().plot(kind='barh')
V7	Which mode of transport is the cheapest on average?	groupby('Primary_Mode')['One_Way_Cost_INR'].mean().plot(kind='barh')

#	Question / Analysis Performed	Python Technique
V8	(Horizontal Bar Chart) How satisfied are students with their commutes overall? (Bar Chart)	df['Satisfaction'].value_counts().sort_index().plot(kind='bar')

SUMMARY – KEY FINDINGS AT A GLANCE

Metric	Finding
Total Responses	724 students
Most Used Mode	Metro (18.09%)
Fastest Mode (avg)	Metro ~39 min
Cheapest Mode (avg)	Metro ~Rs. 23
Most Expensive Mode	Cab (Ola/Uber) ~Rs. 298
Average Travel Time	~57 minutes
Average One-Way Cost	~Rs. 127
Biggest Bottleneck	Waiting for Bus/Metro (133 cases)
Most Common Starting Point	Kothrud (129 students)
Missing Values	132 in Delay_Points — filled with 'None'

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